

Heri Purnawan

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EDUCATION

Institut Teknologi Sepuluh Nopember

Surabaya, Indonesia

Dr. in Applied Mathematics

August 2020 – August 2024

- Thesis title: *Disturbance Observer-based Stochastic Model Predictive Control Design for Systems with Parameter Uncertainties and Disturbances*
- Supervisor: Prof. Subchan, S.Si., M.Sc., Ph.D
- Co-Supervisors: Dr. Tahiyatul Asfihani, S.Si., M.Si.
Prof. Seungkeun Kim, Ph.D (Aerospace Engineering, Chungnam National University)

Institut Teknologi Sepuluh Nopember

Surabaya, Indonesia

M.Si. in Applied Mathematics

August 2016 – August 2018

- Thesis title: *Disturbance Compensating Model Predictive Control for Warship Heading Control in Missile Firing Mission*
- Supervisors: Subchan, S.Si., M.Sc., Ph.D
Dr. Dieky Adzkiya, S.Si., M.Si.

Institut Teknologi Sepuluh Nopember

Surabaya, Indonesia

S.Si. in Applied Mathematics

August 2011 – August 2015

- Thesis title: *Linear Quadratic Regulator (LQR) for Flight Stability of LSU-05*
- Supervisors: Dr. Dra. Mardlijah, MT
Ir. Eko Budi Purwanto, MT (National Institute of Aeronautics and Space (LAPAN))

PUBLICATIONS

Journal Articles

- International
 1. **H. Purnawan**, T. Asfihani, S. Kim, and S. Subchan, "Model predictive control design under stochastic parametric uncertainties based on polynomial chaos expansions for f-16 aircraft," *Journal of Robotics and Control (JRC)*, vol. 5, no. 3, pp. 723–732, 2024.
 2. N. Ilmayasinta and **H. Purnawan**, "Optimal control in a mathematical model of smoking," *Journal of Mathematical & Fundamental Sciences*, vol. 53, no. 3, 2021.
 3. **H. Purnawan**, S. Subchan, D. Adzkiya, and T. Asfihani, "Trajectory tracking of coordinated multi-robot systems using nonlinear model predictive control," *Nonlinear Dynamics and System Theory*, vol. 21, no. 3, pp. 303–314, 2021.
- National
 1. B. A. Iza, **H. Purnawan**, M. K. Huda, and N. Laili, "Design of fuzzy logic controller based on Gustafson-Kessel clustering for quadcopter altitude control," *CAUCHY: Jurnal Matematika Murni dan Aplikasi*, vol. 11, no. 01, pp. 1234-1250, 2026.
 2. **H. Purnawan**, A. R. Wakhid, Q. A. Fiddina, B. A. Iza, T. M. Sanusi, and S. Cahyaningtias, "Uncertainty-aware Kalman filtering via intrusive polynomial chaos for disturbance estimation," *ZERO: Jurnal Sains, Matematika dan Terapan*, vol. 9, no. 03, pp. 991-1003, 2025.
 3. **H. Purnawan**, "Real-time differentiation from noisy data using a sliding mode-based method with low-pass filtering," *Fidelity: Jurnal Teknik Elektro*, vol. 7, no. 02, pp. 141-149, 2025.
 4. R. A. Faroh, **H. Purnawan**, and M. M. Junaidi, "Tingkat depresi, stres, dan kecemasan mahasiswa teknik elektro selama pembelajaran daring," *Tarbiyah Wa Ta'lim: Jurnal Penelitian Pendidikan dan Pembelajaran*, vol. 10, no. 1, pp. 1–11, 2023.
 5. **H. Purnawan**, R. A. Faroh, M. H. Wahyudi, and S. Cahyaningtias, "Optimal control with treatment and water sanitation for cholera epidemic model," *EKSAKTA: Berkala Ilmiah Bidang MIPA*, vol. 23, no. 03, pp. 373–381, 2023.

6. **H. Purnawan**, U. Ilmi, R. A. Faroh, and F. Resmi, "Moving horizon state estimation with application to autonomous vehicle," *Inprime: Indonesian Journal of Pure and Applied Mathematics*, vol. 5, no. 1, pp. 47–59, 2023.
7. **H. Purnawan** and S. Subiono, "Barisan aritmetika bertingkat dengan menggunakan interpolasi lagrange," *Limits: Journal of Mathematics and Its Applications*, vol. 19, no. 2, pp. 145–154, 2022.

Conference Proceedings

- International

1. **H. Purnawan**, T. Asfihani, S. Kim, and S. Subchan, "Stochastic model predictive control for flight path angle tracking of f-16 aircraft based on polynomial chaos," in *2024 International Seminar on Intelligent Technology and Its Applications (ISITIA)*, IEEE, 2024, pp. 599–604.
2. S. Subchan, L. R. Amalia, T. Asfihani, and **H. Purnawan**, "Reference tracking of quadrotor using modified nonlinear model predictive control based on nonlinear disturbance observer," in *International Conference on Mathematics: Pure, Applied and Computation*, Springer, 2023, pp. 87–104.
3. U. Ilmi and **H. Purnawan**, "Study on the problems of particle dynamics and rigid bodies (mechanical engineering field) and electrical circuits (electrical engineering)," in *Proceeding International Conference on Environment Health, Socioeconomic and Technology*, vol. 1, 2022, pp. 213–226.
4. **H. Purnawan**, T. Asfihani, and S. Subchan, "Disturbance observer model predictive control with application to uav pitch angle control," in *AIP Conference Proceedings*, AIP Publishing, vol. 2641, 2022.
5. **H. Purnawan**, U. Ilmi, R. A. Faroh, and A. B. A. A. Rizqi, "Positioning estimation of autonomous car using extended kalman filter," in *Proceeding International Conference on Environment Health, Socioeconomic and Technology*, vol. 1, 2022, pp. 96–102.
6. **H. Purnawan**, T. Asfihani, D. Adzkiya, et al., "Disturbance compensating model predictive control for warship heading control in missile firing mission," in *Journal of Physics: Conference Series*, IOP Publishing, vol. 1108, 2018, p. 012 035.
7. **H. Purnawan**, T. Asfihani, D. Adzkiya, et al., "Mathematical modeling and simulation of missile firing impact force on warship," in *2018 International Seminar on Research of Information Technology and Intelligent Systems (ISRITI)*, IEEE, 2018, pp. 566–570.
8. T. Asfihani, D. Adzkiya, D. M. Rosyid, **H. Purnawan**, R. Kamilah, et al., "Estimation of the corvette sigma motion in missile firing mission," in *2017 5th International Conference on Instrumentation, Control, and Automation (ICA)*, IEEE, 2017, pp. 203–207.
9. **H. Purnawan**, E. B. Purwanto, et al., "Design of linear quadratic regulator (lqr) control system for flight stability of lsu-05," in *Journal of Physics: Conference Series*, IOP Publishing, vol. 890, 2017, p. 012 056.

- National

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RESEARCH FUNDINGS

Research for Early-Career Lecturers (PDP)	2026
<i>Ministry of Higher Education, Science, and Technology (KEMENDIKTISAINTEK)</i>	<i>Indonesia</i>
• Principal Investigator (PI) • Research Title: Analisis Batas Efektivitas Polynomial Chaos pada Kalman Filter untuk Estimasi Gangguan Sistem Pesawat Tempur (<i>Analysis of the Effectiveness Limits of Polynomial Chaos on the Kalman Filter for Disturbance Estimation in Fighter Aircraft Systems</i>)	
Research for Early-Career Lecturers (PDP)	2025
<i>Ministry of Higher Education, Science, and Technology (KEMENDIKTISAINTEK)</i>	<i>Indonesia</i>

- Principal Investigator (PI)
- Research Title: Pengembangan Kalman Filter untuk Estimasi Gangguan pada Pesawat Tempur dengan Ketidakpastian Parameter dan Noise Data Sensor
(*Development of Kalman Filter for Disturbance Estimation in Fighter Aircraft with Uncertainty Parameters and Sensor Data Noise*)

EXPERIENCES

Universitas Negeri Surabaya <i>Department of Data Science</i>	September 2025 – Present <i>Surabaya, Indonesia</i>
• Lecturer • Research: <i>Data-driven for dynamical systems</i>	
Universitas Islam Lamongan <i>Department of Electrical Engineering</i>	February 2022 – August 2025 <i>Lamongan, Indonesia</i>
• Lecturer • Research: <i>System Modelling and Control</i>	
Indonesia College <i>Math Tutor</i>	2016 – 2018 <i>Surabaya, Indonesia</i>
Institut Teknologi Sepuluh Nopember <i>Department of Mathematics</i>	2016 – 2017 <i>Surabaya, Indonesia</i>
• Lecturer Assistant • Calculus 1 & 2	
Junior High School 1 Sugio <i>Mathematical Olympiad Tutor</i>	2015 – 2016 <i>Lamongan, Indonesia</i>
National Institute of Aeronautics and Space (LAPAN) <i>Aeronautics Technology Center</i>	2014 <i>Bogor, Indonesia</i>
• Internship Program • Project title: <i>Modeling and Simulation of LAPAN Surveillance UAV (LSU)-05</i>	

ACHIEVEMENTS

Indonesian Education Scholarship (BPI) <i>Doctoral Study Completion</i>	2023 <i>Ministry of Education, Culture, Research and Technology</i>
Postgraduate Scholarship for Researcher (BPUP) <i>Doctoral's Degree</i>	2020 – 2022 <i>Institut Teknologi Sepuluh Nopember</i>
Graduated Cumlaude <i>Master's Degree</i>	2018 <i>Institut Teknologi Sepuluh Nopember</i>
Fresh Graduate Scholarship <i>Master's Degree</i>	2016 – 2018 <i>Institut Teknologi Sepuluh Nopember</i>
Graduated Cumlaude <i>Bachelor's Degree</i>	2015 <i>Institut Teknologi Sepuluh Nopember</i>
Bidik Misi Scholarship <i>Bachelor's Degree</i>	2011 – 2015 <i>Ministry of Education and Culture</i>

TEACHING

Universitas Negeri Surabaya <i>Spring Semester</i>	September 2025 – Present
• 4920203007 Advanced Calculus, Dept. of Data Science, 2026 • 4920203004 Probability Theory, Dept. of Data Science, 2026 • 4920203009 Data Structure and Algorithm, Dept. of Data Science, 2026 • 4920203035 Digital Image Processing, Dept. of Data Science, 2026 • 4920203086 Machine Learning, Dept. of Data Science, 2026	

Fall Semester

- 4920203003 Basic Programming, Dept. of Data Science, 2025
- 4920202030 Text Processing, Dept. of Data Science, 2025
- 4920203071 Digital Signal Processing, Dept. of Data Science, 2025
- 4920203010 Discrete Mathematics, Dept. of Data Science, 2025
- 4920203008 Matrix Algebra, Dept. of Data Science, 2025

Universitas Islam Lamongan

February 2022 – August 2025

Spring Semester

- TE4485 Engineering Mathematics II, Dept. of Electrical Engineering, 2025
- TE2215 Fundamentals of Chemistry, Dept. of Electrical Engineering, 2022 – 2025
- TE1205 Fundamentals of Computer and Programming, Dept. of Electrical Engineering, 2024
- TE7251 Artificial Intelligence, Dept. of Electrical Engineering, 2022 – 2024
- IFMWP03 Linear Algebra, Dept. of Informatics Engineering, 2022

Fall Semester

- TE3233 Fundamentals of Control System, Dept. of Electrical Engineering, 2022 – 2024
- TE4249 Linear System, Dept. of Electrical Engineering, 2022 – 2024
- TE5278 Numerical Method, Dept. of Electrical Engineering, 2022 – 2024
- KL1209 Fundamentals of Calculus, Dept. of Environmental Health, 2024
- KKM24420 Ordinary Differential Equation, Dept. of Mathematics Education, 2024
- KPKK311 Engineering Mathematics/Calculus, Dept. of Civil Engineering, 2022 – 2023

SKILLS & RESEARCH INTERESTS

Skills

- Languages: *Indonesia, English*
- Coding: *Matlab, C, C++, Python*
- Misc.: *Training, consultation, LATEX typesetting*

Research Interests

- Differential Equation
- Numerical Analysis
- Optimization
- Estimation
- Optimal Control
- Model Predictive Control
- System Identification
- Uncertainty Quantification

REFERENCES

Prof. Subchan

Department of Mathematics, Institut Teknologi Sepuluh Nopember

Professor

Surabaya, Indonesia

- Address: Building F, 2nd Floor, ITS Campus, Keputih, Sukolilo-Surabaya 60111
- E-mail: subchan@its.ac.id

Dr. Dieky Adzkiya

Department of Mathematics, Institut Teknologi Sepuluh Nopember

Assistant Professor

Surabaya, Indonesia

- Address: Building F, 2nd Floor, ITS Campus, Keputih, Sukolilo-Surabaya 60111
- E-mail: dieky@its.ac.id

Prof. Mardlijah

Department of Mathematics, Institut Teknologi Sepuluh Nopember

Professor

Surabaya, Indonesia

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- E-mail: mardlijah@its.ac.id